

E-Commerce Behavioral Analytics: Abnormal User Detection & Implicit Item Property Prediction

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1. Executive Summary

E-commerce platforms generate enormous volumes of implicit behavioral data — clicks, views, cart additions, purchases — but this data arrives noisy, unlabeled, and only indirectly connected to what a business actually needs to know: *what does this customer want, and can we trust this traffic signal in the first place?*

This project addresses both questions through two connected modules built on the Retailrocket e-commerce clickstream dataset (2.75M+ real anonymized events):

- **Abnormal User Detection** — identifies and removes bot-driven and anomalous browsing traffic before it can bias downstream analytics or recommendation systems.
- **The Algorithm** — uses the cleaned behavioral data to predict the implicit category of items a visitor is likely to add to their cart, based solely on their prior viewing sequence.

Key outcomes:

- Flagged and removed traffic from **7.4% of visitors**, eliminating **25.7% of noisy view events**, while explicitly protecting every visitor who ever converted (added to cart or purchased) from being misclassified as abnormal.
- Built a two-stage prediction pipeline achieving **94.4% Top-1** and **97.2% Top-3 accuracy** on implicit category prediction — validated with leakage-safe, time-based evaluation rather than a naive random split.
- Demonstrated a validation-first approach throughout: synthetic ground-truth injection, three-way cross-model agreement checks, and explicit baseline comparisons at every stage, rather than reporting a single headline number in isolation.

Why this matters for the business: noisy, bot-contaminated data silently degrades recommendation quality and invalidates A/B testing results, while accurate implicit-intent prediction is the foundation of personalization — the difference between a generic storefront and one that anticipates what a customer is looking for before they say it.

2. Introduction

2.1 E-Commerce Analytics in Context

Modern e-commerce platforms operate on behavioral signals that are almost entirely implicit. A customer rarely tells a platform "I want a mid-range laptop under a certain budget" — instead, they browse, hover, add items to a cart, abandon some, and purchase others. Extracting reliable signal from this sequence of actions is the foundation of modern recommendation systems, personalization engines, and demand forecasting.

2.2 Why Recommendation Systems Matter

Recommendation quality directly drives revenue in e-commerce — through cross-sell, upsell, and reduced search friction. But a recommendation system is only as good as the behavioral data it is trained on. If that data is contaminated with automated, non-human traffic, the resulting recommendations reflect bot behavior patterns rather than genuine customer intent.

2.3 Challenges in Raw E-Commerce Datasets

- **Implicit feedback only** — no explicit ratings or stated preferences, only inferred intent from actions
- **Severe class imbalance** — views vastly outnumber cart additions, which vastly outnumber purchases
- **Noisy/bot traffic** — automated scraping and monitoring traffic can constitute a significant share of raw logs
- **High cardinality** — thousands of item categories and millions of unique visitors, most with very sparse individual histories
- **Time-varying attributes** — item properties (including category) can change over time, requiring care to avoid using future information when reconstructing historical behavior

2.4 Problem Statement

Given raw, unlabeled clickstream data, this project addresses:

1. How can abnormal (bot-like) traffic be identified and removed without an available ground-truth label, and without harming genuine converting customers?
2. Once cleaned, can a visitor's *implicit* intent — specifically, which product category they will add to cart — be predicted from their browsing sequence alone?

3. Dataset Overview

Source: [Retailrocket E-commerce Dataset, Kaggle](#) — real, anonymized behavioral data collected from an e-commerce website.

File	Description	Scale (as used in this project)
events.csv	Visitor event log: timestamp, visitorid, event (view/addtocart/transaction), itemid, transactionid	2,756,101 events across 1,407,580 unique visitors
item_properties_part1.csv	Time-varying item attributes (property-value pairs per item per timestamp), part 1 of 2	Large-scale property log; filtered to categoryid property for this project
item_properties_part2.csv	Continuation of item property log, part 2 of 2	Combined with part 1 for full category history
category_tree.csv	Hierarchical category structure (categoryid, parentid)	Category hierarchy reference

Event type distribution (raw):

Event	Count	Share
view	2,664,312	96.7%
addtocart	69,332	2.5%
transaction	22,457	0.8%

This distribution illustrates the core challenge of implicit feedback modeling in e-commerce: the events that matter most commercially (addtocart, transaction) are a small fraction of total activity, and any useful model must be built primarily from view events acting as weak, noisy signals of interest.

Data-quality challenges identified:

- Raw event data included a meaningful share of abnormal/bot-like traffic requiring dedicated detection and removal (Module 2)
- Item categories are time-varying, requiring timestamp-aware joins rather than static lookups (addressed in Module 1)
- ~25.6% of events had no resolvable category information at the time they occurred and were excluded from category-dependent modeling to avoid introducing incorrect labels

4. Business Understanding

4.1 Why Predicting Implicit Item Properties Matters

Customers rarely specify product attributes explicitly. A model that can infer the category a visitor is trending toward — before an explicit search or cart action — enables proactive recommendation, more relevant category-page ordering, and better targeting of promotional content.

4.2 Why Abnormal User Detection Is Business-Critical

- **Recommendation system integrity** — bot traffic introduces browsing patterns (extremely high volume, low diversity, robotic timing) that do not reflect real customer preference. Left uncleaned, these patterns bias collaborative filtering and content recommendation models.
- **Split-test (A/B test) validity** — abnormal users add statistical noise and bias to experiment results; a test that appears to show a meaningful lift or drop in conversion may be an artifact of bot contamination rather than genuine user response.
- **Infrastructure and cost impact** — bot traffic consumes serving and analytics infrastructure without any corresponding business value.

4.3 Real-World Applications

- Personalized homepage and category-page ranking
 - Real-time "customers who viewed this also viewed" recommendation widgets
 - Fraud and scraping detection for competitive intelligence protection
 - Cleaner analytics dashboards for merchandising and inventory teams
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5. Data Preprocessing & Cleaning

Applied across both modules:

- **Timestamp conversion** — raw millisecond epoch timestamps converted to proper datetime objects for chronological operations
- **Type enforcement** — `visitorid` and `itemid` cast to consistent integer types to prevent silent join failures
- **Session reconstruction** — a 30-minute inactivity rule was used to segment each visitor's raw event stream into discrete sessions, which underlies several session-level behavioral features
- **Leakage-safe temporal joins** — item category information was joined to events using an "as-of" backward join (`merge_asof`), ensuring that only category information genuinely known at each event's timestamp was used, rather than a static/global lookup that could leak future category reassignments backward in time
- **Neutral imputation** — for visitors lacking sufficient activity to compute certain behavioral features (e.g., click velocity, timing regularity), the population median was imputed rather than a placeholder extreme value, ensuring such visitors are treated as "typical" on that dimension by default rather than artificially forced toward an anomalous score

- **Event filtering** — after abnormal-user detection, only `view` events belonging to confirmed abnormal visitors were removed; all `addtocart` and `transaction` events were preserved unconditionally, since these represent confirmed real business value regardless of the associated visitor's browsing pattern elsewhere
 - **Target cardinality reduction** — for category prediction, the top 30 most frequent target categories (selected using only the training partition, to avoid leakage) were retained, with all remaining categories grouped into a single "other" bucket
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6. Exploratory Data Analysis (EDA)

Key behavioral insights derived from the cleaned event log:

- **Event funnel is steep:** `view` → `addtocart` → `transaction` shows roughly a 40:1 drop from view to addtocart, and a further ~3:1 drop from addtocart to transaction — consistent with typical e-commerce funnels, where browsing intent only rarely converts to a cart action
- **Category "stickiness" dominates browsing behavior:** the large majority of visitors who view a category also add-to-cart from that same category — a simple "same as last viewed category" heuristic already achieves 94.4% accuracy, underscoring how habitual real browsing behavior tends to be
- **A visible minority of visitors deviate from their recent browsing category** (~5.5% of add-to-cart events) — this smaller group is precisely where a predictive model has the opportunity to add genuine value over a naive heuristic
- **Traffic includes a detectable abnormal-behavior segment:** a subset of visitors display high-volume, low-diversity, and unusually regular (low-variance) click timing — patterns consistent with automated rather than human browsing, and clearly separable from typical visitor behavior in behavioral feature space (visualized via PCA projection)
- **Session and timing features separate cleanly** between "normal" and "abnormal" segments post-detection — the abnormal segment shows markedly higher median event counts and unique-item counts alongside near-zero conversion, consistent with non-human browsing rather than simply "highly engaged" real customers

Visualizations used: PCA scatter plot (2D projection of behavioral feature space, colored by anomaly flag), feature importance bar charts (for both the deviation classifier and category classifier), confusion matrix heatmap (Stage A deviation classifier), event-mix bar chart (before vs. after cleaning), and a contamination-parameter sensitivity line plot (synthetic recall vs. real-user flagging rate).

7. Module: The Algorithm

7.1 Objective

Predict the product category a visitor will add to cart, using only features derivable from their `view` activity prior to that action — addressing the core implicit-feedback challenge that customer intent is never stated explicitly.

7.2 Relationship Between View and Add-to-Cart Events

Every `addtocart` event was matched to the visitor's most recent `view` history using a backward as-of join, so that only information genuinely available *before* the cart action is used as a feature — this is what makes the evaluation a legitimate test of predictive power rather than hindsight.

7.3 Feature Engineering Strategy

Feature	Description
<code>last_viewed_categoryid</code>	Most recent category the visitor viewed before the cart action
<code>second_last_viewed_categoryid</code>	One step further back — gives the model a sense of browsing trajectory
<code>category_revisit_count</code>	Cumulative count of times the visitor has revisited their current category
<code>cum_views,</code> <code>cum_unique_categories</code>	Running totals capturing browsing breadth and depth
<code>cum_mode_categoryid</code>	The visitor's most frequently viewed category so far (running mode)

7.4 Model Development Pipeline

1. **Input features:** behavioral features listed above (categorical + numeric)
2. **Target variable:** item category associated with the `addtocart` event (top-30 categories + "other" bucket)
3. **Train-test split:** strict time-based split (80/20 by event timestamp) — not a random split, to prevent the model from being evaluated on data chronologically earlier than its training set
4. **Model selection:** LightGBM, chosen for its efficiency on large tabular data with mixed categorical/numeric features and native handling of categorical encodings
5. **Architecture — two-stage design:**
 - **Stage A (binary classifier):** predicts whether the visitor will *deviate* from their last-viewed category at all (~5.2% of training cases do)
 - **Stage B (multiclass classifier):** trained exclusively on the subset of cases where deviation occurs, predicting *which* category the visitor moves to
 - Final prediction defaults to the last-viewed category unless Stage A triggers Stage B — concentrating model capacity on the minority of genuinely hard cases rather than diluting it across the majority of already-easy cases
6. **Hyperparameter tuning:** RandomizedSearchCV with TimeSeriesSplit cross-validation for Stage B, tuning `num_leaves`, `max_depth`, `learning_rate`, `min_child_samples`, `subsample`, `colsample_bytree`, and regularization terms

7.5 Why This Architecture Was Suitable

A single-stage multiclass model trained on the full dataset would spend the majority of its learning capacity re-deriving "predict last-viewed category" for the ~95% of cases where that is already correct. Splitting the problem into "will this visitor deviate?" and "if so, to where?" allows each sub-model to specialize, and makes the model's behavior directly interpretable: performance can be diagnosed separately for the deviation-detection step and the category-assignment step.

7.6 Model Performance

Model	Top-1 Accuracy
Majority-class baseline	71.4%
Same-as-last-viewed baseline	94.42%
Two-stage pipeline (final)	94.39%
Additional metric	Value
Top-3 Accuracy	97.24%
Top-5 Accuracy	97.92%
Macro F1-score	0.852
Stage A (deviation classifier) — Accuracy	94.68%
Stage A — Precision	65.52%
Stage A — Recall	10.12%

Interpretation:

- The final Top-1 accuracy is marginally below the naive "same-as-last-viewed" baseline — reported transparently rather than adjusted, since the baseline itself is genuinely strong (browsing behavior is highly habitual).
 - Where the two-stage architecture demonstrates real value is in **Top-3 and Top-5 accuracy**, and in **Macro F1** — both reflect the model's ability to correctly rank the true category highly even in the harder ~5.5% of cases where visitors change category, which a single-baseline heuristic cannot address at all by construction.
 - Stage A's precision (65.5%) meaningfully exceeds its recall (10.1%), indicating the model is conservative — when it does flag a likely deviation, it is right roughly two-thirds of the time, but it under-flags the full set of true deviations. This is a deliberate, interpretable trade-off rather than an unexamined limitation: given that false "deviation" triggers actively risk degrading otherwise-correct baseline predictions, a conservative Stage A protects overall accuracy.
 - ROC-AUC was not the primary evaluation metric for this multiclass problem; Top-K accuracy and Macro F1 were used instead as more appropriate measures for a ranked, imbalanced multiclass target.
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8. Module: Abnormal User Detection

8.1 Definition of Abnormal Users

Visitors whose behavioral signature (extreme activity volume, mechanically regular timing, systematic catalog-scanning patterns, or zero-conversion despite high engagement) is inconsistent with typical human browsing, and consistent instead with automated agents, scrapers, or monitoring bots.

8.2 Business Impact of Bot Traffic

Left unaddressed, bot traffic inflates apparent engagement metrics, distorts item-popularity signals feeding into recommendation systems, and introduces bias into any split-test that isn't explicitly filtered for it.

8.3 Feature Generation Strategy

Nine behavioral feature groups were engineered per visitor:

Group	Features	Rationale
Volume	Total events, views, add-to-carts, transactions	Bots often exhibit extreme activity volume
Diversity	Unique items viewed, item diversity ratio	Distinguishes broad browsing from narrow/no engagement
Conversion	View-to-cart ratio, cart-to-transaction ratio	Genuine customers show meaningfully different conversion patterns than scrapers
Velocity	Average inter-click gap, views per minute	Superhuman click speed is a strong bot signal
Timing Regularity	Coefficient of variation of click gaps	Human timing is naturally variable; scripted timing is often suspiciously constant
Repeated-Item Ratio	Ratio of repeat views to unique views	Distinguishes exploratory browsing from repetitive/looping bot behavior
Session-Based	Number of sessions, average/max session duration, events per session	Captures engagement structure over time
Night-Activity	Proportion of activity during 12am–5am	A secondary, weaker signal for automated/monitoring traffic
Sequential Catalog-Scan Score	Spearman correlation between view order and item ID	Detects systematic, sequential scraping through the catalog — a signature unlikely in genuine human browsing

8.4 Model Building Process

Primary model: Isolation Forest — selected for its efficiency on high-dimensional, large-scale tabular data (1.4M+ visitors) and its suitability for unsupervised anomaly detection without requiring labeled bot examples.

Threshold (contamination) selection strategy: Since no ground-truth bot labels exist in the raw dataset, 500 synthetic bot profiles were constructed in raw feature space with an unambiguous bot signature (high volume, high click rate, robotic timing regularity, systematic scanning, zero conversion) and injected into the feature matrix. The `contamination` parameter was then selected as the smallest value that still recovered at least 95% of these synthetic bots — balancing detection sensitivity against the risk of over-flagging genuine users purely to inflate recall.

Independent validation (avoiding single-model bias):

- **Local Outlier Factor (LOF)**, a density-based method conceptually distinct from Isolation Forest's tree-based approach, was run on a 50,000-visitor sample: **92.1% agreement** with Isolation Forest's flags.
- **Rule-based conjunction method** — a deliberately strict, interpretable rule requiring multiple independent suspicious signals simultaneously — showed **45.1%** of its flagged users also caught by Isolation Forest, indicating the model is capturing genuinely high-confidence cases rather than arbitrary noise.

Business-rule override: any visitor who ever completed an `addtocart` or `transaction` event was excluded from the final abnormal-user list regardless of model score — reflecting the business priority that mistakenly removing a real, converting customer is a materially worse outcome than leaving a marginal bot in the dataset.

8.5 Feature Importance & Behavioral Separation

Post-detection comparison of feature medians between normal and abnormal segments showed clear, behaviorally meaningful separation (e.g., abnormal visitors showing several times higher total events and unique items viewed, alongside near-zero conversion), confirming the flagged segment reflects a genuine distinct behavioral pattern rather than a statistical artifact.

9. Model Evaluation

9.1 The Algorithm — Evaluation Summary

	Metric	Value
Top-1 Accuracy		94.39%
Top-3 Accuracy		97.24%
Top-5 Accuracy		97.92%

Metric	Value
Macro F1-score	0.852
Cross-validation (Stage B, macro F1, <code>RandomizedSearchCV</code>)	0.173

Note: the Stage B cross-validation macro F1 (0.173) reflects performance on the genuinely hard multiclass subset only (deviating cases, 31 classes) — it is not directly comparable to the blended pipeline's overall Macro F1 (0.852), which includes the much larger, easier "no deviation" majority.

9.2 Abnormal User Detection — Evaluation Summary

Metric	Value
Synthetic bot recall (at selected contamination)	100%
Silhouette Score	0.689
Isolation Forest vs. LOF agreement	92.1%
Isolation Forest vs. rule-based overlap	45.1%
Final abnormal users flagged (post business-override)	7.4%
View events removed	25.7%

Note: standard supervised classification metrics (Precision/Recall/ROC-AUC against ground truth) are not applicable here, since no true bot labels exist for this dataset — validation instead relies on synthetic-recall and independent cross-method agreement, as detailed in Section 8.4.

9.3 Bias–Variance and Error-Type Discussion

- **The Algorithm:** Stage A's low recall (10.1%) relative to its precision (65.5%) reflects a deliberate bias toward conservative predictions — the cost of a false positive (incorrectly overriding an otherwise-correct baseline prediction) was judged higher than the cost of a false negative (missing a genuine category change), given the strength of the baseline being overridden.
- **Abnormal User Detection:** the business-rule override explicitly accepts a higher false-negative rate for bot detection (some borderline bots may remain unflagged if they ever converted) in exchange for a near-zero false-positive rate on real customers — a deliberate business-driven trade-off rather than an oversight.

10. Results & Findings

- Real browsing behavior is highly habitual: a simple "same as last viewed category" heuristic is a surprisingly strong baseline (94.4%), and the genuine modeling opportunity lies specifically in the smaller subset of visitors who change intent mid-session.

- Behavioral signals alone (without any content-based item features) are sufficient to detect abnormal traffic with high cross-method agreement, indicating these features capture a genuine, distinguishable behavioral pattern rather than noise.
 - Protecting converting visitors via business-rule override is essential — a purely model-driven anomaly threshold would have incorrectly flagged tens of thousands of real, revenue-generating visitors.
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11. Business Impact

- **Improved recommendation input quality** — removing ~25.7% of bot-driven view events directly reduces the noise feeding into any downstream collaborative-filtering or content-based recommendation system trained on this data.
 - **More reliable experimentation** — cleaned traffic reduces the risk of bot contamination biasing A/B test outcomes.
 - **Foundation for proactive personalization** — the category-prediction model demonstrates that implicit intent signals, even in the hardest ~5.5% of "changed their mind" cases, can be captured well enough (97%+ Top-5 accuracy) to inform real-time recommendation re-ranking.
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12. Challenges Faced

- **Large-scale data handling** — processing 2.75M+ events and item-property files spanning multiple large CSV files required chunked reading and careful memory management
 - **No ground-truth labels** — for abnormal user detection, requiring a validation strategy built from synthetic injection and cross-method agreement rather than standard supervised metrics
 - **Severe class imbalance** — both in event types (views vastly outnumber conversions) and in category deviation (only ~5.5% of add-to-cart events represent a genuine category change)
 - **Time-varying item attributes** — requiring leakage-safe, timestamp-aware joins rather than simple static lookups
 - **Avoiding subtle temporal leakage** — an earlier iteration of the category-prediction pipeline selected its top-N target categories using the full dataset (train + test combined); this was identified and corrected to use the training partition only
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13. Future Improvements

- **Deep learning sequence models** (e.g., GRU/Transformer-based session encoders) to capture longer, non-linear browsing sequences beyond the current last-two-category feature window
 - **Real-time anomaly detection** via streaming feature computation, rather than the current batch-based approach
 - **Graph-based recommendation** leveraging the category hierarchy (`category_tree.csv`) more directly, beyond simple category-ID features
 - **Hybrid recommendation systems** combining the implicit-intent signal from this project with content-based item similarity
 - **Scalable cloud deployment** (e.g., batch scoring pipelines) for production-scale application of both modules
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14. Technologies Used

Category	Tools
Language	Python
Data Manipulation	Pandas, NumPy
Machine Learning	Scikit-learn, LightGBM
Anomaly Detection	Isolation Forest, Local Outlier Factor
Visualization	Matplotlib, Seaborn
Statistical Analysis	SciPy
Environment	Jupyter Notebook

15. Conclusion

This project delivers an end-to-end e-commerce behavioral analytics solution addressing two interconnected, business-critical problems: ensuring the integrity of raw behavioral data through rigorous, multi-method-validated abnormal user detection, and extracting genuine predictive value from that cleaned data through a carefully architected two-stage implicit-intent model. Rather than reporting isolated headline metrics, this project consistently benchmarks results against meaningful baselines, cross-validates findings across independent methods, and transparently documents where model performance is strong and where it is limited — reflecting the validation discipline expected of production-grade data science work rather than leaderboard-style optimization alone.

16. References

- Retailrocket E-commerce Dataset, Kaggle:
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